**Unit 2 Topic 4: All Definitions**

| Term | Definition |
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| **Biodiversity** | **The variety of living organisms in an area, encompassing species, genetic, and habitat diversity.** |
| **Endemism** | **The ecological state of being unique to a defined geographic location and found nowhere else.** |
| **Species** | **A group of organisms capable of interbreeding and producing fertile offspring.** |
| **Habitat** | **The specific environment where an organism lives, characterized by physical conditions and other species present.** |
| **Species richness** | **The total number of different species present within a habitat or ecosystem.** |
| **Genetic diversity** | **The number and frequency of alleles present within the gene pool of a species or population.** |
| **Niche** | **The role or function of a species within its habitat, including its interactions and use of resources.** |
| **Adaptation** | **An inherited feature (anatomical, behavioural, physiological) that increases an organism's chance of survival.** |
| **Natural selection** | **The process by which individuals with advantageous traits survive and reproduce more, passing on these traits.** |
| **Evolution** | **A change in allele frequencies within a population over generations.** |
| **Speciation** | **The formation of new and distinct species through evolutionary processes.** |
| **Allopatric speciation** | **Speciation resulting from geographic isolation preventing gene flow between populations.** |
| **Sympatric speciation** | **Speciation occurring within the same geographic area due to reproductive isolation.** |
| **Reproductive isolation** | **Mechanisms (behavioural, ecological, mechanical, temporal, genetic) preventing two species from interbreeding.** |
| **Population bottleneck** | **An event causing a significant reduction in a population's size, greatly reducing genetic diversity.** |
| **Genetic drift** | **Random changes in allele frequency within a population, often significant in small populations.** |
| **Gene flow** | **The transfer of genetic material (alleles) between populations through migration and reproduction.** |

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| **Inbreeding depression** | **Reduced biological fitness resulting from mating between closely related individuals, increasing harmful alleles.** |
| **Heterozygosity index** | **A measure of genetic diversity calculated as the proportion of heterozygotes in a population.** |
| **Simpson's Diversity Index** | **A numerical measure of biodiversity accounting for species richness and evenness in a habitat.** |
| **Classification** | **Organising living organisms into hierarchical groups based on shared characteristics and evolutionary relationships.** |
| **Three-domain system** | **A classification system dividing life into three groups: Bacteria, Archaea, and Eukarya.** |
| **Cellulose** | **Polysaccharide consisting of β-glucose monomers; primary component of plant cell walls.** |
| **Starch** | **A polysaccharide energy storage molecule in plants, consisting of amylose and amylopectin.** |
| **Lignin** | **A rigid polymer in plant cell walls, providing structural support and waterproofing.** |
| **Xylem** | **Plant tissue transporting water and mineral ions; composed of dead, lignified cells.** |
| **Phloem** | **Plant tissue transporting organic solutes (sucrose); made of living sieve tubes and companion cells.** |
| **Sclerenchyma** | **Supportive plant tissue composed of dead, lignified cells without transport function.** |
| **Sustainability** | **Meeting present needs without compromising the ability of future generations to meet theirs.** |
| **Micropropagation** | **Cloning plants using tissue culture techniques to produce genetically identical individuals.** |
| **Cryopreservation** | **Preservation of biological material at very low temperatures (–196°C using liquid nitrogen).** |
| **Seed bank** | **Facility for storing seeds under controlled conditions to preserve genetic diversity and viability.** |
| **Artificial pollination** | **Manual transfer of pollen to ensure controlled fertilisation and genetic diversity.** |
| **Zoos** | **Facilities housing animals in controlled environments for conservation, breeding, and public education.** |

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| **Botanical garden** | **Controlled environment for cultivating diverse plant species for conservation, research, and education.** |
| **Arboretum** | **Botanical collection specialising in the cultivation and conservation of trees.** |
| **In situ conservation** | **Conservation of species in their natural habitats.** |
| **Ex situ conservation** | **Conservation of species outside their natural habitats (e.g. zoos, seed banks, botanical gardens).** |
| **Stud book** | **Detailed records tracking lineage, genetic traits, and breeding history of captive organisms.** |
| **Artificial insemination** | **Direct introduction of sperm into a female's reproductive system without natural mating.** |
| **In vitro fertilisation** | **Fertilisation of an egg by sperm in laboratory conditions; embryos are implanted into females.** |
| **Gene bank** | **Repository for storing genetic material (sperm, eggs, embryos, seeds) to conserve genetic diversity.** |
| **Reintroduction** | **Release of captive-bred individuals into their natural habitats to re-establish populations.** |